

Unsupervised Learning

What is Unsupervised Learning?

Unsupervised Learning is a type of Machine Learning where the data does not contain labels or correct outputs.

The model learns patterns, structures, and relationships from the data by itself.

Key Idea

Supervised Learning, learn with answers.

Unsupervised Learning, find hidden patterns without answers.

Goal of Unsupervised Learning

Find hidden structure in data

Group similar data points (Clustering)

Reduce dimensions while preserving important information (Dimensionality Reduction)

Detect anomalies/outliers

Discover associations and similarities

Common Techniques in Unsupervised Learning

Clustering

Grouping similar data points.

Example:

Customers with similar shopping behavior.

Dimensionality Reduction

Reducing number of features while keeping important information.

Example:

PCA (Principal Component Analysis)

Anomaly Detection

Finding unusual or abnormal data points.

Example:

Fraud detection.

Why Use Unsupervised Learning?

Explore hidden patterns

Understand data structure

Reduce complexity

Detect outliers

Improve supervised learning models

Compress data efficiently

Use Cases of Unsupervised Learning

Use Case

Purpose

Customer

Group customers by

Segmentation

behavior

Market Segmentation Group products/markets

Use Case	Purpose
Fraud Detection	Detect unusual activity
Recommendation Systems	Find similar users/products
Image Compression	Reduce image size
Medical Analysis	Discover disease patterns
PCA Preprocessing	Improve supervised models

Supervised vs Unsupervised Learning

Supervised	Unsupervised
Data has labels	No labels
Correct outputs known	Only input data available

Supervised

Learns predictions

Classification,
Regression

Unsupervised

Learns patterns

Clustering, PCA, Anomaly
Detection

K-Means Clustering

K-Means is a clustering algorithm that groups similar data points into K clusters using centroids.

Main Idea

We divide data into groups based on similarity.

Each cluster has a center point called:

Centroid

Steps of K-Means

1. Decide Number of Clusters

Example:

$k = 4$

2. Initialize Centroids

Randomly choose center points.

3. Assign Clusters

Assign each data point to nearest centroid using Euclidean Distance.

Euclidean Distance Formula

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

4. Reassign Centroids

Calculate new center points.

5. Repeat

Assign clusters again until centroids stop changing.

Important Point

K-Means is:

Centroid-Based Clustering Algorithm

Elbow Method

A method used to find the optimal K value in K-Means clustering.

Graph

- X-axis → Number of Clusters (K)
- Y-axis → WCSS

WCSS => Inertia

Within Cluster Sum of Squares

Measures how far data points are from their centroid.

Formula Idea

$$d_1^2 + d_2^2 + d_3^2 + \dots + d_n^2$$

Lower WCSS means tighter clusters.

Main Logic

As K increases:

- WCSS decreases

But after some point:

- improvement becomes small

That turning point forms an “elbow”.

Best K value is selected there.

K-Means Disadvantages

1. Need to Choose K

K value must be manually initialized.

2. Sensitive to Outliers

Outliers can shift centroids.

3. Poor for Non-Circular Data

K-Means assumes circular/spherical clusters.

For irregular shapes:

Use DBSCAN

DBSCAN

Full Form

Density-Based Spatial Clustering of Applications with Noise

A density-based clustering algorithm that groups closely packed points and identifies outliers as noise.

Key Features

1. Density-Based Algorithm

Clusters are formed based on data density.

2. Non-Parametric

No need to define number of clusters beforehand.

3. Epsilon Distance (ϵ)

Defines neighborhood radius around points.

Advantage Over K-Means

Handles:

- Outliers
- Non-circular clusters

better than K-Means.

Dimensionality Reduction

Reducing the number of input features while preserving important information.

Why Need Dimensionality Reduction?

Large dimensions create:

Curse of Dimensionality

Curse of Dimensionality

Problems

- Hard to find patterns
- Distance-based algorithms fail

- Higher storage requirements
- Slower training
- Increased complexity

Algorithms Affected

Distance-based algorithms:

- KNN
- K-Means
- DBSCAN

struggle in high dimensions.

Goal

Reduce dimensions while keeping maximum useful information.

Methods of Dimensionality Reduction

1. Feature Selection

Selecting the most important existing features.

Example:

Keep useful columns, remove unnecessary ones.

2. Feature Extraction

Creating new compressed features from old features.

Example

Engine Width + Engine Height



Transform & Extract



Engine Size

Important Point

Some information loss may occur,
but prediction can still work effectively.

PCA (Principal Component Analysis)

PCA is a dimensionality reduction technique that transforms high-dimensional data into lower dimensions while preserving maximum variance.

Main Goal of PCA

Find a new direction (Principal Component) where:

Maximum variance is captured.

Principal Component

A new transformed feature that captures maximum information from data.

PCA Logic

Original features may be correlated.

PCA combines them into fewer powerful dimensions.

Benefits of PCA

- Reduces dimensions
- Speeds up training
- Reduces storage
- Removes redundancy
- Improves visualization
- Helps reduce noise

Important Connection

High-dimensional data



Curse of dimensionality



Distance-based models struggle



Need dimensionality reduction



Use:

- Feature Selection
- Feature Extraction



Most powerful extraction method:

PCA

Final Full Connection

Unsupervised Learning



No labels available



Need to discover hidden patterns



Use Clustering



K-Means groups data using centroids



Need best K value



Use Elbow Method



K-Means struggles with irregular clusters/outliers



Use DBSCAN



High-dimensional data creates complexity



Need Dimensionality Reduction



Feature Selection / Feature Extraction



Most popular technique:

PCA

Final Summary

Unsupervised Learning allows machines to discover hidden structures in unlabeled data using techniques like clustering, anomaly detection, and dimensionality reduction, where algorithms such as K-Means, DBSCAN, and PCA help organize, simplify, and understand complex datasets efficiently.